

Multi-turn Conversation Chatbot

1. Project Title

Building Chatbot using Gemini GenAI API

2. Project Overview

This project focuses on designing, developing, and deploying a production-ready chatbot application powered by the Google Gemini GenAI API.

The chatbot must be built for a specific domain and deployed in a cloud environment following real-world AI engineering standards.

The goal is to move beyond a basic demo and implement a scalable, secure, and well-structured GenAI application.

3. Domain Selection

You may choose any of your interested domain or one domain from the following:

- E-commerce Chatbot
- Blog Assistant Chatbot
- Mental Health Support Chatbot
- News Chatbot
- Career Advisor Chatbot
- Financial Advisor Chatbot

The chatbot must provide domain-specific, structured, and intelligent responses.

4. Technical Requirements

The solution must follow production-grade AI architecture and best practices.

4.1 Gemini API Integration

- Integrate Google Gemini GenAI API
- Secure API key management using environment variables
- Structured request and response handling
- Proper exception handling and fallback mechanism
- Logging of API calls and errors
- Token usage optimization

Expectation: API logic must be modular and separated from UI code.

4.2 Advanced Prompt Engineering

The chatbot must demonstrate strong prompt engineering skills:

- Structured system prompts
- Role-based instructions
- Domain-specific constraints

Prompts must be configurable and reusable.

4.3 User Interface

Develop an interactive UI using:

- Streamlit

UI must include:

- Chat-style interface
- Real-time response rendering
- Conversation history display
- Loading indicator
- User-friendly layout